

ES 670 Assignment

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Q.1)

a) My hometown is Aligarh (Uttar Pradesh). Based on proximity and availability of data under the Continuous Ambient Air Quality Monitoring System (CAAQMS), the three nearest CPCB monitoring stations are:

- Kalindi Kunj, Khurja – UPPCB
- Yamunapuram, Bulandshahr – UPPCB
- Omex Eternity, Vrindavan – UPPCB

For detailed analysis, the Yamunapuram, Bulandshahr station was selected due to availability of consistent and complete data.

b) Daily average data for the years 2019, 2020, and 2021 was collected from CAAQMS portal for the selected station as shown in excel. The variables considered include:

- PM_{2.5} concentration ($\mu\text{g}/\text{m}^3$)
- Relative Humidity (RH, %)
- Ambient Temperature (AT, $^{\circ}\text{C}$)
- Wind Speed (WS, m/s)
- Wind Direction (WD, degrees)

c) The time series plot of PM_{2.5} concentrations reveals several important patterns

- **Strong Seasonal Periodicity:** The data shows a definitive annual cycle where PM_{2.5} levels surge during winter (Q4–Q1) and dissipate during the monsoon and summer months (Q3), typical of meteorological washout and inversion effects.
- **Extreme Peak Intensity:** Winter pollution peaks consistently exceed $400 \mu\text{g}/\text{m}^3$, with a notable historical maximum surpassing $500 \mu\text{g}/\text{m}^3$ in November 2020—nearly 10x the recommended safe limits.
- **Baseline Floor Stability:** Despite extreme spikes, the summer troughs remain relatively stable between $20\text{--}50 \mu\text{g}/\text{m}^3$. However, 2021 shows a slightly higher baseline floor than 2020, suggesting a slow increase in background ambient pollution due to partial resumption of human activities.
- **2020 Lockdown Signature:** Mid-2020 exhibits an exceptionally stable and low-noise period compared to 2019, likely reflecting the drastic reduction in industrial and vehicular emissions during that year's lockdowns.

- **Pre-Peak Warning Signals:** Small increases in PM_{2.5} levels are observed in late October each year. These act as early physical indicators of the massive atmospheric stagnation and smog episodes that follow in November.

d)

Correlation Analysis					
	PM _{2.5} (µg/m ³)	RH (%)	AT (°C)	WS (m/s)	WD (°)
PM _{2.5} (µg/m ³)	1				
RH (%)	-0.02821	1			
AT (°C)	-0.58960	-0.42374	1		
WS (m/s)	-0.10870	-0.09547	-0.06999	1	
WD (°)	0.08008	-0.10244	-0.10899	0.09473	1

- **Strong Negative Correlation with Temperature:** PM_{2.5} shows a significant negative correlation with Ambient Temperature (AT) of -0.5896. This confirms the "winter effect" seen in your previous plot; as temperatures drop, PM_{2.5} concentrations rise sharply due to reduced mixing heights and temperature inversions.
- **Weak Influence of Relative Humidity:** The correlation between PM_{2.5} and Relative Humidity (RH) is nearly negligible at -0.0282. This suggests that in this specific dataset, humidity levels are not a primary driver for particulate concentration changes compared to temperature.
- **Inverse Relationship with Wind Speed:** There is a negative correlation of -0.1087 with Wind Speed (WS). This indicates that higher wind speeds facilitate the dispersion and "flushing" of pollutants, while stagnant air (low WS) leads to accumulation, though the effect here is relatively modest.
- **Negligible Impact of Wind Direction:** The correlation with Wind Direction (WD) is very low at 0.0800. This suggests that pollution levels in this study area are likely driven more by local regional stagnation and temperature-induced trapping rather than a specific directional source or "long-range transport."
- **Meteorological Inter-dependence:** Ambient Temperature and Relative Humidity are moderately negatively correlated (-0.4237). This means that during colder periods (when PM_{2.5} is high), the air is generally less humid, which might influence the chemical secondary formation of particulates.
- **Dominance of Thermal Forcing:** Comparing all variables, Ambient Temperature is the single most dominant meteorological predictor for PM_{2.5} in this dataset. Its correlation coefficient is nearly 6 times stronger than that of Wind Speed, highlighting that thermal stability is the primary factor controlling air quality here.

Q.2)

a) The data was cleaned by removing null values and outliers. For the lockdown analysis, the IQR method applied ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) and the upper bound for outliers was determined to be 156.96 µg/m³, and the lower bound was 0 as the value was coming out to be negative which is not possible.

b) A single-factor ANOVA was performed to determine if there were significant differences in mean PM_{2.5} concentrations across eight defined periods (Pre-Lockdown, Lockdowns 1-6, and Post-Lockdown)

SUMMARY				
<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>
Pre-Lockdown	64	5901.28	92.2075	1196.05988
Lockdown 1	21	1011.11	48.1480952	467.851326
Lockdown 2	19	1036.3	54.5421053	415.59714
Lockdown 3	13	648.2	49.8615385	177.863697
Lockdown 4	15	1008.21	67.214	922.11684
Lockdown 5	30	1390.93	46.3643333	172.80515
Lockdown 6	31	895.09	28.873871	165.377845
Post-Lockdown	61	2155.03	35.3283607	516.493421

Hypotheses:

- H₀: Mean PM_{2.5} concentrations are equal across all periods
- H₁: At least one period has a different mean

ANOVA						
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	139151.958	7	19878.8512	32.9982102	3.7087E-32	2.04692328
Within Groups	148195.837	246	602.422103			
Total	287347.796	253				

Based on observed trends in the dataset, F-value of 32.99 significantly exceeded the F-critical value of 2.04. With a P-value of 3.7087E-32, the test confirms a statistically significant difference in mean PM_{2.5} concentrations across the different lockdown phases.

c) Following the one-way ANOVA, Tukey's Honest Significant Difference (HSD) test was performed as a post-hoc analysis to identify specific pairs of time periods that exhibit statistically significant differences in mean PM_{2.5} concentrations.

While ANOVA determines the presence of an overall significant difference among group means, it does not indicate which particular groups differ. Tukey's test addresses this by conducting pairwise comparisons between all lockdown phases while controlling for Type I error. The results indicate that:

- Pre-Lockdown mean (115.36 µg/m³) is significantly different from every other period. All 7 comparisons exceed their HSD thresholds (diff range: 48–87 µg/m³). Confirms that pre-COVID pollution was dramatically higher.
- Lockdown 4 mean (67.21 µg/m³) is significantly higher than Lockdown 6 mean = 28.87 and Post-Lockdown (mean = 35.33). This is notable — Lockdown 4 shows a rebound possibly due to partial resumption of activities.

- No significant differences among Lockdown 1,2,3,5,6 and Post Lockdown (means range 28.87–54.54 $\mu\text{g}/\text{m}^3$). They form a statistically homogeneous cluster confirming sustained $\text{PM}_{2.5}$ reduction during lockdown.
- Pre-Lockdown and Lockdown 4 are the two outlier periods. All remaining lockdown phases and Post-Lockdown are statistically equivalent, showing that COVID-19 restrictions reduced $\text{PM}_{2.5}$ by 51–75% and this effect persisted post-lockdown.